

**RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
ACADEMIC YEAR 2024-2025**

FACULTY AWARDS, ACHIEVEMENTS AND HONOURS RECEIVED

GUEST LECTURE DELIVERED

- Mr. R. Arun Kumar, AP(SG)/Mech served as one of the resource persons for the **"Two-day Menor - Mentee Program"** organized by Francis Xavier Engineering College, Tirunelveli during 04-05 July, 2024. 16 faculty members from 6 different institutions attended the program.



- Mr. R. Arun Kumar, AP(SG)/Mech and Master Trainer, EDII, Govt. of TN served as a resource person as well as jury for the one-day **"Ideation Camp"** organized by EDII, Govt. of Tamil Nadu held on 07 Aug., 2024 at Collectorate, Sivagangai district. Students and mentors of 25 shortlisted teams from various engineering and polytechnic colleges of Sivagangai, Pudukottai, and Ramnad districts attended the event.



- Mr. R. Prabhakaran, AP/Mech delivered the guest lecture on the topic of “**Recent Developments in Additive Manufacturing**” held at P.A.C.Ramasamy Raja Polytechnic College, Rajapalayam on 12 Aug., 2024.



- Dr. S. Maharajan, ASP-I/Mech visited **Mando Automotive India Pvt Ltd, Chennai** for initiation of training for operating engineer on 23 Aug., 2024.



- Mr. R. Arun Kumar, AP(SG)/Mech and IGBC Accredited Professional, served as the resource person for the "**Seminar on Energy Conservation**" organized by the RIT-IGBC Chapter in association with the Department of Civil Engineering on 30 Aug., 2024.
- Mr. R. Arun Kumar, AP(SG)/Mech and Master Trainer of Entrepreneurship Development and Innovation Institute (EDII) served as the resource person for the three-day Boot Camp of "**EDII Hackathon**" held at SCAD College of Engineering and Technology, Tirunelveli between the 04 Sep., and 06 Sep., 2024.



- Mr. R. Arun Kumar, AP(SG)/Mech and Convener, RIT-IIC gave an **“Orientation to the I Year CSBS”** department students about the function of IIC and its contribution to strengthening the innovation ecosystem. The session was held on 05 Sep., 2024.

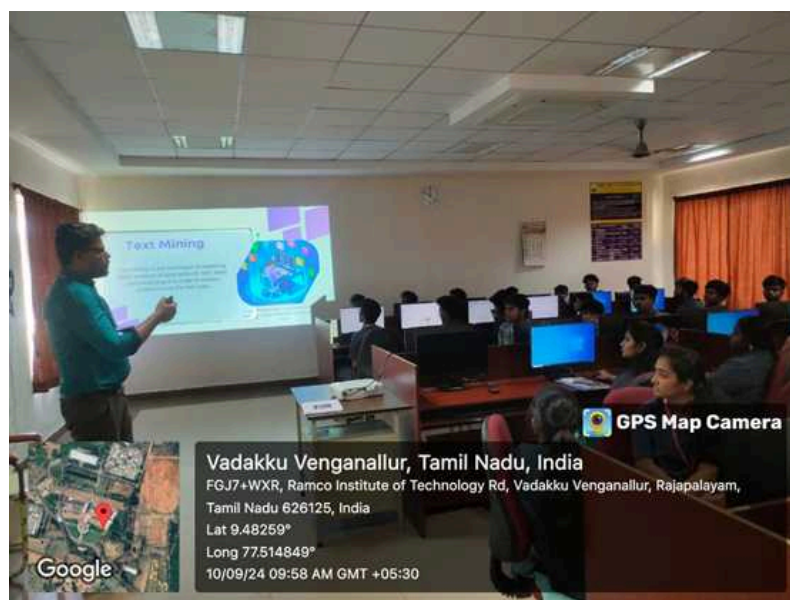


- Dr. G. Prabu ram, AP(SG)/Mech acted as a resource person for the session on **"Acceleration and Incubation Opportunities for Students and Faculties - Early**

Stage Entrepreneurs" held on 09 Sep., 2024 for the students of Mechanical Engineering.



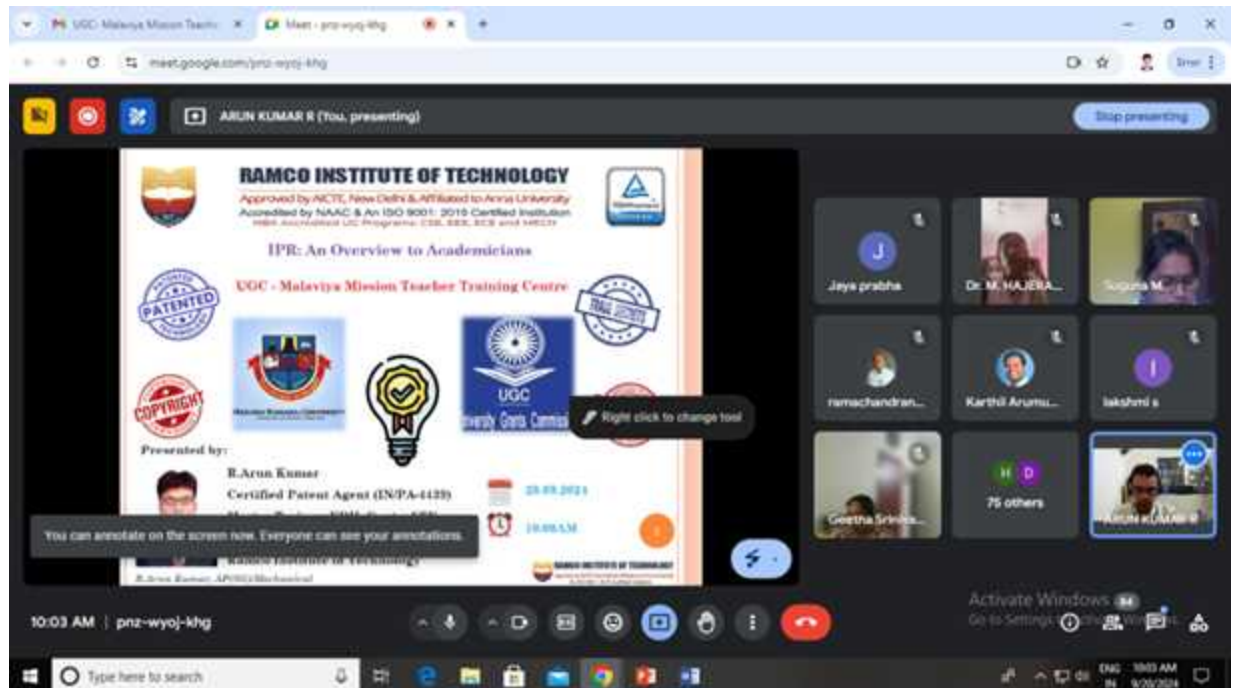
- Mr. S. Valai Ganesh, AP(SG)/Mech delivered the guest lecture on “**Natural Language Programing and Hands-on-Training**” to the III Year CSBS students on 10 Sep., 2024 at CSBS Laboratory.



- Mr. L. Karthikeyan, AP(SG)/Mech, delivered the guest lecture on "**Modern Mobility - Accelerating through the past**" held at PAC Ramasamy Raja Polytechnic College, Rajapalayam on 11 Sep., 2024.



- Mr. R. Arun Kumar, AP(SG)/Mech and Certified Patent Agent (IN/PA-4439) delivered his expertise on the topic "**IPR: An overview for Academicians**" during the "**Refresher Course in Commerce and Management**" organized by UGC - Malaviya Mission Teacher Training Centre, Madurai Kamaraj University, Madurai on 20 Sep., 2024. 85 faculty members from various institutions attended the event.



- RIT IIC organized a session titled "**Safeguard your Innovation: An Introduction to IPR**" on 19 Oct., 2024. Mr. R. Arun Kumar, AP(SG)/Mech and Certified Patent Agent (IN/PA-4439) acted as resource person for the session.



- Mr. S. Valai Ganesh, AP(SG)/Mech delivered a guest lecture on "**Mechatronics and IoT**" in JP College of Engineering, Tenkasi on 07 Nov., 2024.



- Mr. R. Arun Kumar, AP(SG)/Mech acted as a resource person for the "**Workshop on IP Rights: A Tool to Protect Intellectual Property**" organized by Department of CSBS, KGiSL Institute of Technology, Coimbatore on 20 Nov., 2024.



- RIT-Research Council and RIT Institution's Innovation Council organized a "**Half Day Workshop on Protect Your Innovations: A Comprehensive Guide to IPR**" on 09 Dec., 2024. Mr. R. Arun Kumar, AP(SG)/Mech, and Dr. G. Prabu ram, AP(SG)/Mech were the resource person for the Workshop. The workshop focused on IPR Awareness practice, patent search and draft



- Mr. R. Arun Kumar, AP(SG)/Mech and Certified Patent Agent (IN/PA-4439) served as a resource person for the session titled **"IP Rights and IP Management for Start-ups"** during One-week ATAL Faculty Development Program on **"Enhancing Education through Design Thinking and Entrepreneurial Mindset"** organized by SRM TRP Engineering College, Trichy on 16 Dec., 2024.



- Mr. R. Arun Kumar, AP(SG)/Mech and Certified Patent Agent (IN/PA-4439) served as a resource person for the session titled "**Seminar on IP Strategies for Aviation Management**" organized by AJK College of Arts and Science, Coimbatore on 17 Dec., 2024.



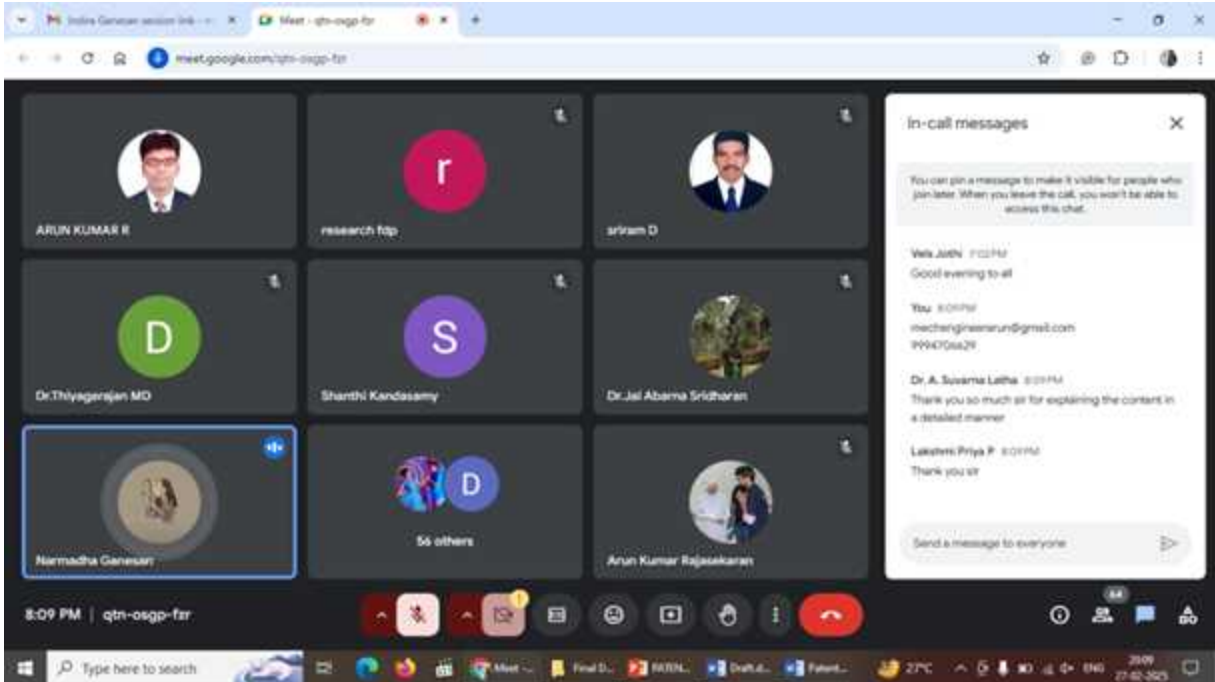
- Mr. R. Arun Kumar, AP(SG)/Mech delivered a seminar titled "**IPR and its importance for Innovators and Entrepreneurs**" organized by the department of CSE, K.Ramakrishnan College of Engineering, Trichy on 07 Feb., 2025.



- Mr. S. Valai Ganesh, AP(SG)/Mech delivered a guest lecture on “**Role of robotics in healthcare sector**” in IEI Tuticorin Local Center on 19 Feb., 2025.



- Mr. R. Arun Kumar, AP(SG)/Mech was invited as a resource person for the Faculty Development Programme on Intellectual Property Rights organized by Indra Ganesan Institutions, Trichy. He delivered his thoughts on the title "**Patent Drafting - Do's and Don'ts**" on 27 Feb., 2025.



- Dr. G.Prabu ram, ASP-I/Mech, delivered a lecture for Engineering Knowledge sharing on "**Cp and Cpk calculation**" organized by Rajapalayam Mills Private Limited to all unit supervisors on 05 Mar., 2025.



- Mr. R. Arun Kumar, AP(SG)/Mech and Certified Patent Agent (IN/PA-4439) delivered his insights to the industrial as well as academic participants during the Awareness Programme titled “**Navigating IP rights: A guide for MSMEs**” organized by Francis Xavier Engineering College, Tirunelveli on 07 Mar., 2025.



- Mr.R.Arun Kumar, AP(SG)/Mech delivered his expertise for the students of the Department of IT on the topic "**From Ideas to Action: Building a Product**" organized by the Department of IT in association with RIT IIC on 10 Mar., 2025.



- Institution's Innovation Council of N.M.S Sermathai Vasan College for Women, Madurai, organized a Seminar on "**Basics of IPR and its Importance for Innovators and**

Entrepreneurs" on 17 Mar., 2025. Mr. R. Arun Kumar, AP(SG)/Mech acted as a resource person for the session and shared his expertise with 60+ participants.



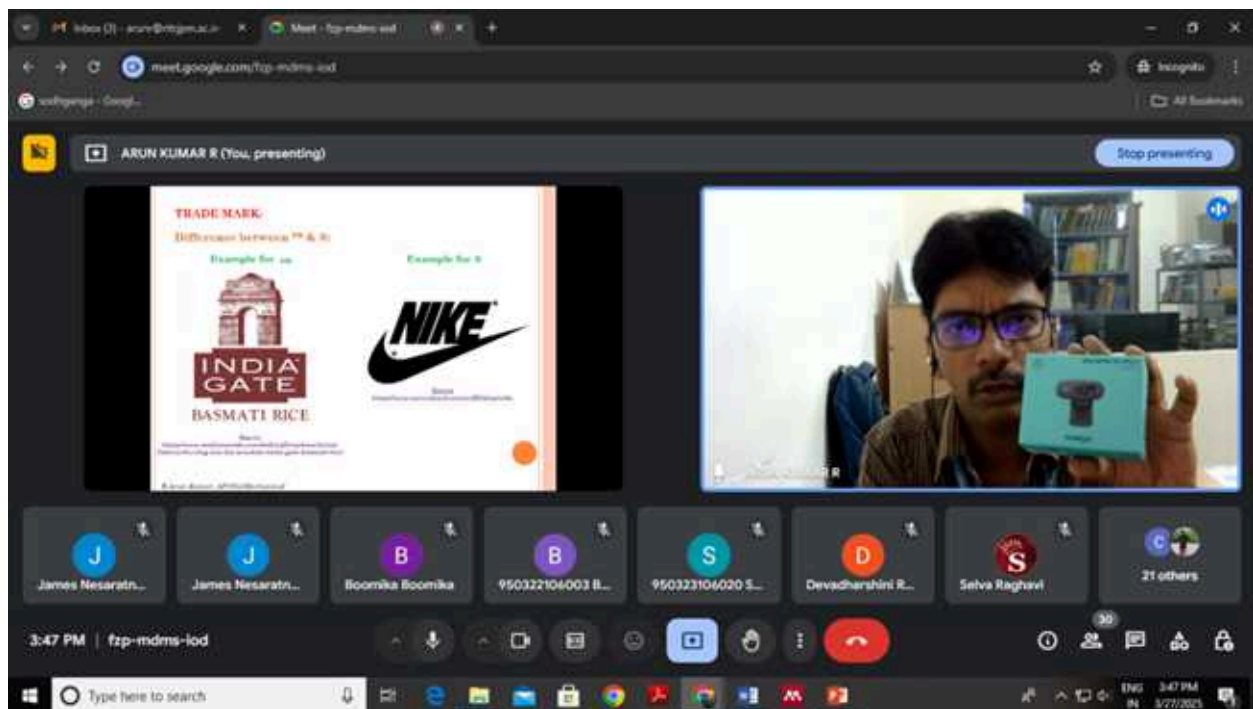
- Mr. R. Arun Kumar, AP(SG)/Mech and Certified Patent Agent served as a resource person for the Workshop titled "**Copyright Registration – A Comprehensive Guide for Creators**" organized by RIT - IIC for the students of III Year CSBS on 20 Mar., 2025.



- Dr. G. Prabu ram, ASP-I/Mech, delivered a lecture on "**Internet of Things and Applications**" organized by the Department of Mechanical Engineering In association with Institution of Engineers(India) at P.A.C. Ramasamy Raja Polytechnic College on 27 Mar., 2025.



- Mr. R. Arun Kumar, AP(SG)/Mech shared his thoughts as a resource person for the online Webinar on "**IP Protection and Management for Start-ups**" organized by Grace College of Engineering on 27 Mar., 2025.



- Dr. R.Venkatesh, AP(III)/Mech delivered the Industry Institute Lecture series, topic on "**Design of Bearings**" in Rajapalayam Mills Ltd., Rajapalayam on 27 Mar., 2025.



- Dr. G. Prabu ram, ASP-I/Mech, delivered a lecture on National Level workshop on the title "**ISmart Sensing: A Hands on adventure with Arduino and IOT for Young makers**" organized by the Department of Mechanical Engineering for school students at Ramco Institute of Technology College on 28 Mar., 2025. Hands-on session was Engaged by Mr.M.Manojkumar & Mr.G.Kathirvel of II Year CSBS.



- Mr. S. Valai Ganesh, AP(SG)/Mech, delivered a lecture on National Level workshop on the title "**Robotics and Automation: Industry 5.0 & Smart Factory**" organized by the Department of Mechanical Engineering for school students at Ramco Institute of Technology College on 28 Mar., 2025.



- Dr. G. Prabu ram, ASP I/Mech, delivered a Guest lecture on "**Introduction to IPR**" organized by IETE students chapter ECE for III year ECE students, Ramco Institute of Technology on 12 May, 2025.



- Dr. G. Prabu ram, ASP I/Mech, delivered a Guest lecture on "**Entrepreneurship Development Programme**" organized by EDII TN Virudhunagar District for VKP (Valnthu Kattuvom Project) Beneficiaries on 08 May, 2025.



CHIEF GUEST

- Dr. S. Rajakarunakaran, Prof.&Head/Mech invited as a Mock Audit member for “**NAAC Accreditation**” at SRM TRP Engineering College, Trichy during 28-29 Sep., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech selected as a Mechanical Divisional Committee Member for IE(I) Tuticorin Local Center for the year 2024-2026 and attended 21st Annual General Meeting at Bell Hotel, Tuticorin on 22 Oct., 2024.



- Mr. S. Valai Ganesh, AP(SG)/Mech attended “**12th Board of Studies Meeting**” at Francis Xavier Engineering College on 23 Dec., 2024.



- Mr. S. Valai Ganesh, AP(SG)/Mech attended “**10th PAQIC Meeting**” for the department of Electrical and Electronics Engineering of Ramco Institute of Technology, Rajapalayam on 09 Jan., 2025.
- Dr. S. Rajakarunakaran, Prof.&Head/Mech/National Executive Committee Member attended “**ISTE National Convention and Yuva Kaushal Utsav 2025**” organised at Lamrin Tech Skills University (LTSU), Punjab during 13-14 Feb., 2025.



- Mr. S. Valai Ganesh, AP(SG)/Mech attended “13th Board of Studies Meeting” at Francis Xavier Engineering College on 30 May, 2025.



REVIEWER

- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**Fake Product Identification using Blockchain Technology**” in Journal of Production Research and Management

- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Experimental Analysis of Line Balancing Issues in Automovite Fuel Tank Bracket Production”** in Journal of Automobile Engineering and Applications on 07 July, 2024
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an articles titled **“identifying malicious/fake users in recommendation systems”**, **“Breast Cancer Detection In Image Processing”** and **“LungSE-Net: Enhanced Lung Cancer Diagnosis via Lightweight CNN Model using Histopathological Images”** in the 7th International Conference on Recent Trends in Image Processing & Pattern Recognition (RTIP2R), Indian Institute of Information Technology, Bhopal, India on 10 July, 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“IP Transcendence: Unifying IPv6 Over IPv4 Network”** in the Journal of Telecommunication, Switching Systems and Networks on 23 July, 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Centre Stage of Wireless Sensors in Healthcare: Application and Remote Patient Management”** in the Recent Trends in Sensor Research and Technology on 23 July, 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Advanced Saline Monitoring System For Medical Precision”** in the Journal of Microelectronics and Solid State Devices on 23 July, 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Wear analysis of Al-SiC-Jute Ash hybrid composite via stir casting process”** in the Journal of Interactions, Springer on 30 July, 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Recycling and Reusing Construction Waste for Sustainable Project Delivery: Practices and Constraints in a Developing Country”** in the Waste and Resource Management (Proceedings of the ICE) on 05 Aug., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“The impact of flipped learning on soft skills for enhanced employability”** in the 4th International Conference on Technology and Innovation in Learning, Teaching and Education taking place on 13 – 15 November 2024 at Khalifa University, Abu Dhabi, United Arab Emirates (UAE) on 03 Sep., 2024.
- Dr. G. Prabu ram, AP(SG)/Mech reviewed an article **“Assessment of Aluminum Alloy 6351 Eggshell Reinforced Composite in Dry Turning Using Response Surface Methodology”** in Journal of Basic and Applied Research International on 07 Sep., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Real-Time Detection of Cracks and Structural Weaknesses in Materials and Surfaces using Improved Deep Layered CNN”** in the Journal of The Institution of Engineers (India): Series C (IEIC), Springer on 18 Sep., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“A Novel Framework for Rice Leaf Disease Detection using Mixed Precision and Grad-CAM”** in the International Conference on Innovations in Data Science organised by Manipal University Jaipur on 19 Sep., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Unlocking Potential: Personalizing Learning and Assessment with Cutting-Edge Technologies”** in the Adopting Artificial Intelligence Tools in Higher Education – Student Assessment, Publisher: CRC Press: Taylor & Francis Group on 25 Sep., 2024.

- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“A Review of Robust Network Security System: Machine Learning for Intrusion Detection”** in the International Conference on Innovations in Data Science organised by Manipal University Jaipur on 30 Sep., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech invited to serve as **“Technical Program Committee member (TPC) for CSNT 2025 (14th IEEE International Conference on Communication Systems and Network Technologies)”** organised by VIT Bhopal University, KotriKalan during 07-08 Mar., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Path and Trajectory Planning of Unmanned Ground Vehicles with Vehicle Dynamic Characters”** in International Journal of Robotics and Automation in Mechanics on 15 Oct., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Programming NFV in 5G Environments”** in 16th International Conference on Computational Intelligence and Communication Networks 2024 is organized by Oriental University, Indore on 17 Oct., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an articles **“Enhanced E-waste Management through Smart Tracking and Monitoring: A Modern Approach to Countering E-waste Management Issues Using A-BiCNN-RNN Model”**, **“A Review on Evolving STEM Education: Balancing Tradition and Innovation for the 21st Century”**, **“Machine Learning Based Indian Stock Market Forecasting”**, **“Technical Analysis and Comparison of Intrusion Detection Systems Using Hybrid Approach”** and **“Social Media Analytics – A tool for Digital Marketing”** in 1st International Conference on Data Computation and Communication is organized by VIT, Bhopal on 19 Oct., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book **“Machine Learning Principles and Techniques”** in Bentham Science Publisher on 21 Oct., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an articles **“Machine Learning-Based Traffic Classification for Dynamic Resource Allocation in IP Networks”**, **“Enhancing Sustainable Agriculture through Smart Architecture: An Adaptive Neuro-Fuzzy Inference System with XGBoost Model”** and **“Interdisciplinary Perspectives on the Convergence of Healthcare, Artificial Intelligence, and Internet of Things”** in 1st International Conference on Data Computation and Communication is organized by VIT, Bhopal on 24 Oct., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Hybrid Energy Storage System for Electric Vehicle”** in Journal of Power electronics system on 08 Nov., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Smart Energy meter using IOT”** in International Journal of Electrical Power and Machine System on 09 Nov., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Electric Vehicle Battery Protection System”** in International Journal of Power Electronics Controllers and Converters on 09 Nov., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an articles **“A Review of Robust Network Security System: Machine Learning for Intrusion Detection”**, **“Advanced Game Design Paradigms: AI as a Catalyst for Modeling Player Dynamics and Experience Refinement”** in International Conference on Innovations in Data Science organized by Manipal University Jaipur on 04 Nov., 2024.

- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book chapter “**Personalized Learning with AI Tailoring Education to Individual Needs**” in Driving Quality Education Through AI and Data Science, IGI Publications on 11 Nov., 2024.
- Dr. G. Prabu ram, AP(SG)/Mech reviewed an article “**Experimental Validation of Cascade Model Predictive Control for Optimizing UAV Quadcopter Stability and Accuracy Using Pixhawk Hardware in Windy Turbulent Mangrove Forest**” in International Journal of Brazilian Society of Mechanical Sciences and Engineering on 17 Nov., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article “**MSM: Point cloud alignment algorithm through multi-scale feature fusion and cross-attention mechanism**” in Intelligent Service Robotics, Springer on 22 Nov., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article “**Real-Time Detection of Cracks and Structural Weaknesses in Materials and Surfaces using Improved Deep Layered CNN**” in Journal of The Institution of Engineers (India): Series C (IEIC), Springer on 23 Nov., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**An IoT based Water-borne Diseases Prediction System**” in Recent Advances in Computer Science and Communications on 05 Dec., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**The Transformative Power of AI and Machine Learning in Education**” in International Journal of Computer Aided Manufacturing on 06 Dec., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book chapter on “**Harnessing AI for Sustainable Futures: Innovations and Challenges**” Bentham Science Publisher on 19 Dec., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**MSM: Point cloud alignment algorithm through multi-scale feature fusion and cross-attention mechanism**” in Intelligent Service Robotics Journal, Springer on 24 Dec., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**Effect of Nanoparticle in Jute/Kenaf/Glass Composite for Interlaminar Shear Strength**” in Interactions Journal, Springer on 04 Jan., 2025.
- Mr. R. Venkatesh, AP/Mech reviewed an article titled “**Surface Evaluation of Recycled Nylon for FDM 3D Printing for the Purpose of Dental Prosthesis Construction**” in Advances in Materials Research, an International Journal on 02 Jan., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book titled “**INFINITE HORIZONS: Unlocking energy sources beyond our planet**” in Bentham Science Publisher on 10 Jan., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**Cooperative Push-Grasping in Clutter via Deep Reinforcement Learning**” in Intelligent Service Robotics, Springer on 20 Jan., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article(s), “**Integrated Framework for Equitable Healthcare AI: Bias Mitigation, Community Participation, and Regulatory Governance**”, “**Satellite Image Analytics for Tree Enumeration for Diversion of Forest Land**”, “**Hands Free Mouse Using Facial Expression for Physically Disabled People**”, “**Marine Mammals Sound Classification Using a Lightweight CNN for Edge Computing Devices**”, “**Design of Stepper Motor Based Dosing System to Enhanced Accuracy & Efficiency for Chemical Industry**”,

“Lightweight CNN for Everyday Human Sound Classification on Edge Devices”, “AI Innovations in Heart Disease Prediction Using Machine Learning and Risk Stratification Using Deep Learning”, “IoT and ML-Based Smart Chair Systems for Health Monitoring and Recommendations” in 14th IEEE International Conference on Communication Systems and Network Technologies organised by VIT Bhopal in association with MIR Labs Gwalior with technical support of IETE Gwalior and IEEE Computer society chapter during 03 Feb., 2025- 08 Feb., 2025.

- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book chapter titled **“Artificial Intelligence for Real Intelligence: Simplifying the basics for everyone’s use in practical life”** published by Bentham Science on 06 Feb., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech invited as a **“Technical Program Committee Member”** for the 2025 IEEE Madhya Pradesh Section Conference (MPCON) on 10 Feb., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article(s), **“Designing and Development a Compliant Platform using 2 Degrees of Freedom Flexure Mechanisms”, “Kinematic Analysis of Wankel Rotary Engine Using Vector Loop Equations”** in International Journal of Computer Aided Manufacturing on 15 Feb., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book on **“Machine Learning for Financial Analytics: Volume 1”** in Benthem Publisher on 15 Feb., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Investigate the Mechanical and Tribological characteristics of Cu- Al₂O₃- Gr composite via stir route”** in Materials Research Journal on 20 Mar., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“A LIME-LSTSNM Approach Based Green Building Sustainability Prediction Using Bim Design”** in Sustainable Computing: Informatics and Systems Journal on 21 Mar., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“A Comprehensive Review on Strength properties of concrete with partial replacement of waste rubber and mineral admixtures”** in Construction Materials (Proceedings of the ICE) Journal on 21 Mar., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article(s) titled **“AgriBotNet: ROS 2-Enabled Automation for Chili Crop Management”, “AI Powered Autonomous Robotic Rover for Pest Management”** in 2025 IEEE 1st International Conference on Materials, Robotics & Automation, Computer, and Control (ICMRACC-2025) organised by the Vimal Jyothi Engineering College, Kannur on 22 Mar., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech acted as a reviewer for the **“2025 IEEE 1st Conference on Materials, Robotics, Automation, Computer and Control (ICMRACC-2025)”** organised by the Vimal Jyothi Engineering College (VJEC), Kerala will be held during 16-18 May, 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book chapter titled **“Intelligent Mobile and IoT Ecosystems - Bridging Cloud, Fog, Edge, and AI”** Publisher: CRC Press: Taylor & Francis Group on 15 Apr., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“A LIME-LSTSNM Approach Based Green Building Sustainability Prediction Using Bim Design”** in Sustainable Computing: Informatics and Systems, Elsevier on 19 Apr., 2025.

- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Cooperative Push-Grasping in Clutter via Deep Reinforcement Learning”** in Intelligent Service Robotics, Springer on 21 Apr., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech invited as a Technical Program Committee Member for the **“17th IEEE International Conference on Computational Intelligence and Communication Networks (CICN 2025)”** is organized by Goa University in Association with IEEE Madhya Pradesh Section and IETE Gwalior on 25 Apr., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled **“Advancements in Humanoid Robot Locomotion: A Review of Control Strategies and Kinematic Models”** in International Journal of Robotics and Automation in Mechanics on 28 Apr., 2025.
- Dr. S. Rajakarunakaran, Prof. and Head/Mech reviewed an article in the Journal of Cleaner Production on 04 May, 2025.
- Dr. V.Sivakumar, Prof./Mech reviewed an article in the Journal of Cleaner Engineering and Technology on 04 May, 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech invited to serve as **“Technical Program Committee Member”** in IEEE International Conference on NExt-Gen Pervasive Computing and Communication 2025 organised by IEEE Nepal during Dec., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an articles titled **“Cloud-Enabled Isolation Forest for Anomaly Detection in UAV-Based Power Line Inspection”, “Reinforcement Learning Driven Smart Charging Algorithms to Enhance Battery Lifespan and Grid Sustainability”** in 5th International Conference on Advancement in Electronics & Communication Engineering 2025 organised by Raj Kumar Goel Institute of Technology, Ghaziabad during Nov., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Cooperative Push-Grasping in Clutter via Deep Reinforcement Learning”** in Intelligent Service Robotics, Springer on 18 May, 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Investigate the Mechanical and Tribological characteristics of Cu- Al₂O₃- Gr composite via stir route”** in Materials Research Journal on 19 May, 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“UAV-Aided User Equipment Access Authentication in 5G/6G Networks”** in 2025 IEEE Madhya Pradesh Section Conference (MPCON) on 19 May, 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article **“Implementation and Analysis of EV Powertrain System on MATLAB Employing PMSM as traction device”** in 5th International Conference on Advancement in Electronics & Communication Engineering 2025 organised by Raj Kumar Goel Institute of Technology, Ghaziabad during Nov., 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an articles **“Advanced Phishing Website Classification Using Ensemble Learning with XGBoost and TabNet”, “DermAI : Skin Disease Prediction Using CNN”, “Seamless Styling: The Integration of AI in Virtual Wardrobes”, “Remote Patient Monitoring System”, “Bike Rental Demand Using Machine Learning: A Data-Driven Approach”** in International Conference on Sustainable Computing and Artificial Intelligence organised by Compucom Institute of Technology & Management, Jaipur during 21-23 May, 2025.

- Dr. V. Sivakumar, Prof./Mech. reviewed an article in the “**Solar Energy Materials and Solar Cells**” on 23 May, 2025
- Dr. V. Sivakumar, Prof./Mech. reviewed an article in the “**Journal of Water Process Engineering**” on 24 May, 2025
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed a book chapter titled “**Remote Patient Monitoring Architectures and Applications**” in the book "Industry 6.0 for Sustainable Supply Chains in Agriculture, Healthcare, and Asset Management" by IGI Publishers on 07 June, 2025.
- Mr. S. Valai Ganesh, AP(SG)/Mech reviewed an article titled “**From Collaborative Robots to Internet of Cobots: Innovations, Applications, and Emerging Trends**” in Journal of Ambient Intelligence and Smart Environments by Sage on 12 June, 2025.

PROPOSAL SUBMISSION

- M. Jeeva, M. Ajay, M. Suriyha, III Mech submitted a proposal titled “**Design and Development of a fine kora grass splitting machine**” in The Institution of Engineers (India) R&D Grant in aid scheme on 05 Nov., 2024. Dr. S. Rajakarunakaran, Prof. and Head/Mech. & Dr. V. Sivakumar, Prof./Mech. Officiated as mentors for this project.
- L. Saravanakumar, V. Suryakumar, IV Mech submitted a proposal “**Smart Waste Segregation: A Sustainable Circular Economy Model**” in Niral Thiruvizha 2.0 under the guidance of Mr. S. Valai Ganesh, AP(SG)/Mech on 23 Jan., 2025.

FACULTY PUBLICATION-ARTICLE/BOOK CHAPTER

- M. Lakshmanan, J. Santhana Krishnan, S. Vignesh and M. Jeeva published a book chapter on “**Experiment of DMLS-Produced SS316L Specimen: Microstructure, Hardness and Corrosion Analysis**”, in Technological Trends in Information Science and Engineering publication, ISBN : 9 7 8-81-9 6 5 0 8 1-5-9, April 2024.
- S. Valai Ganesh, V. Suresh, S. Godwin Barnabas, S. Rajakarunakaran published an article “**Innovative solid waste management strategies for smart cities in Tamil Nadu: challenges, technological solutions, and sustainable prospects**”, Discover Applied Science, 6, 660 (2024). <https://doi.org/10.1007/s42452-024-06404-0>
- S. Maharajan, F. Michael Thomas Rex 2024, Microstructural characterisation studies on hot corrosion behaviour of YSZ/Gd₂Zr₂O₇ based plasma spray coatings on SS316’, Physica Scripta, 100, 015924, Impact Factor: 2.6, DOI 10.1088/1402-4896/ad963d. SCIE & Scopus Indexed
- S. Valai Ganesh, V. Suresh, S. Godwin Barnabas, and S. Rajakarunakaran Published a book chapter (2025) titled “**Advancements in mechatronic systems for manufacturing, robotics, and automation Yaskawa Motoman GP12 robot applications**”, in Mechatronics Concepts, Tools, Applications, and New Trends, Taylor & Francis Group, <https://doi.org/10.1201/9781003494478>
- S. Valai Ganesh, V. Suresh, S. Rajakarunakaran, L. Ganesan, V. Suryakumar (2025) “**Transforming Student Assessments in the Era of Machine Learning: Balancing Innovation and Ethics**”, in Driving Quality Education Through AI and Data Science, IGI Global Scientific Publishing, <https://doi.org/10.4018/979-8-3693-8292-9.ch017>

- V. Suryakumar, S. Valai Ganesh, S. Rajakarunakaran published an article (2025), **“Integrating Advanced Technologies in Mechanical Engineering Education: A Path to Industry 4.0”** in IEI Student Journal, Volume 3, Issue 1, Page No. 101-105.
- S. Valai Ganesh, M. Gomathy Nayagam, V. Suresh, S. Rajakarunakaran, B. Bensujin (2025) **“Adaptive Learning Technologies: Navigating the Road from Hype to Reality”**, in Multimodal Generative AI, Springer, https://doi.org/10.1007/978-981-96-2355-6_4
- S. Godwin Barnabas, K. Arun Vasantha Geethan, **S. Valai Ganesh** (2025), “Commercialization of waste tyre using a Simple Recycling Process”, Environmental Engineering and Management Journal, 24(1), 63-77. <http://doi.org/10.30638/eemj.2025.006>
- Prabu ram, G., Lingadurai, K. Interfacial Behavior of NiP Coatings on Fused Filament-Fabricated 17-4PH: Insights into Corrosion and Erosion. J. of Materi Eng and Perform (2025). <https://doi.org/10.1007/s11665-025-11466-1>. Impact factor: 2.2. SCIE & Scopus Indexed.

AWARD

- Mr. S. Valai Ganesh, AP(SG)/Mech, Team Leader-eLSI along with Dr. M. Gomathy Nayagam, ASP-III/CSBS, Mr. S. Meenakshi Sundravel, AP(SG)/EEE and Mr. B. Kannan, AP(SG)/ECE participated and won Drone- Pluto 1.2 in **“e-Yantra Teachers Robotics Competition National Finals 2024”** organised by IIT Bombay in association with SRM University, Chennai on 20 Aug., 2024.



- Mr. S. Valai Ganesh, AP(SG)/Mech selected for the **“IEI Young Engineer Award 2024”** in Mechanical Engineering discipline. The award will be given at 39th National Convention of Mechanical Engineers to be held at Chandigarh during 13-14 Sep., 2024.
- Dr. V. Sivakumar, Prof./Mech received a grant of Rs.7,500 for the project titled **“Eco-friendly IoT-enabled water purifier”** offered by the Tamil Nadu State Council for

Science and Technology, Chennai under Student Project Scheme 2023-24. The student members associated with the project are M. Aravinthanathan, S. Karan, S. P. Cheran, Batch 2020-2024.

- Mr. S. Valai Ganesh, AP(SG)/Mech received the “**IEI Young Engineer Award 2024**” in Mechanical Engineering discipline conducted during the 39th National Convention of Mechanical Engineers at Chandigarh during 13-14 Sep., 2024.



- Dr. S. Rajakarunakaran, Prof.&Head/Mech received the “**Best Teacher Award (Engineering)**” from the Institution of Engineers India during 57th Engineers Day Celebration in Tamil Nadu State Center, Chennai on 15 Sep., 2024.



- Dr. L. Ganesan, Principal/RIT, Dr. S. Rajakarunakaran, Vice-principal/RIT, Mr. K. Selvaraj, GM(Admin) received the “Institute Excellence Award (Engineering)” from the Institution of Engineers India during 57th Engineers Day Celebration in Tamil Nadu State Center, Chennai on 15 Sep., 2024 .



- Dr. S. Rajakarunakaran, Prof. & Head/Mech has been nominated as “**Research Advisory Board Member**” of P.S.R. Engineering College, Sevalpatti, Sivakasi.
- Dr. G. Prabu ram, AP(SG)/Mech received an award “**Society for the Advancement of Library and Information Science (SALIS)- Library Ambassador**” given by SALIS Chennai on 04 Oct., 2024.
- Mr. S. Valai Ganesh, AP(SG)/Mech elected as a “**Divisional Committee Member-Mechanical Engineering**” for the session 2024-2025 and 2025-2026 in IE(I) Tuticorin Local Center.
- Mr. L. Karthikeyan, AP(SG)/Mech, selected to receive an “**Appreciation Award (Faculty Category)**” at the SAEISS Awards Ceremony held on 19 Oct., 2024, at Hotel Ramada Plaza, Guindy, Chennai, in recognition of his outstanding contributions to SAEIndia Southern Section (SAEISS) during assessment year 2023-2024.



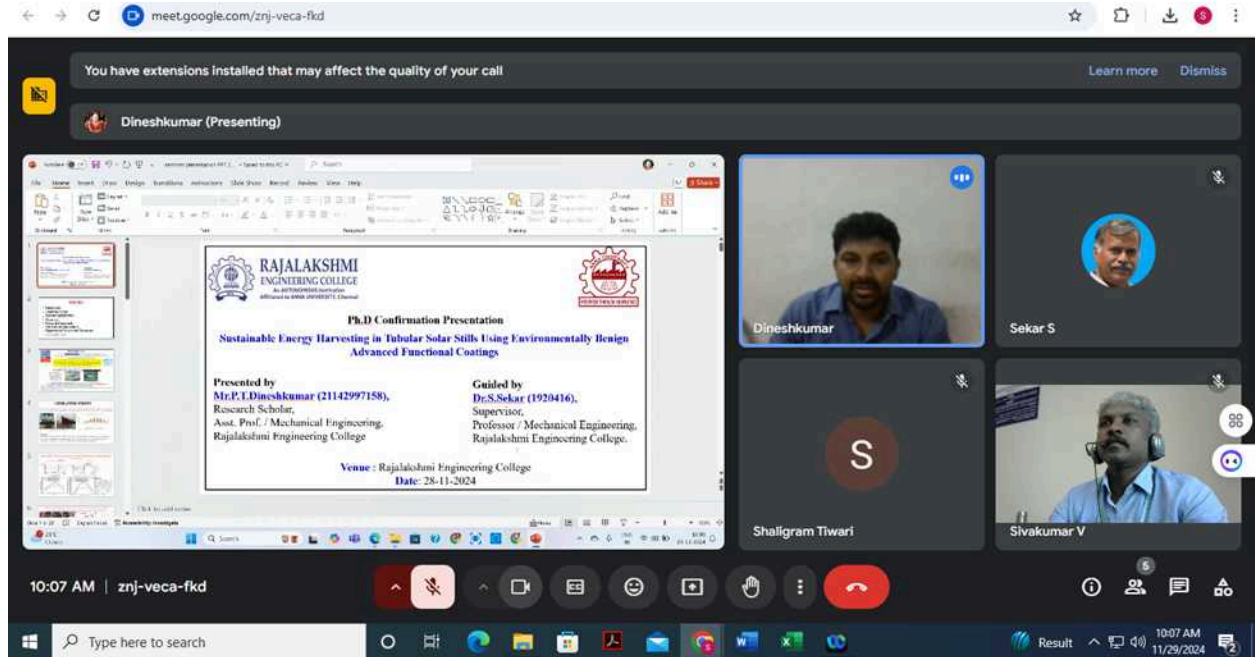
- Mr. R. Arun Kumar, AP(SG)/Mech has been shortlisted for undergoing two months virtual internship by NPTEL for completing the course titled Entrepreneurship and IP Strategy as a topper which was offered by IIT Kharagpur.
- Dr. V. Sivakumar, Prof./Mech. Secured Topper position in an online course "**Applied Thermodynamics for Engineers**" offered by IIT Guwahati via NPTEL.
- Mr. S. Valai Ganesh, AP(SG)/Mech secured Top 2% position with Gold in an online course on "**Automation in Manufacturing**" offered by IIT Guwahati via NPTEL.
- Mr. S. Valai Ganesh, AP(SG)/Mech secured Top 5% position with Silver in an online course on "**Industrial Robotics: Theories for Implementation**" offered by IIT(ISM) Dhanbad via NPTEL
- Mr. L. Karthikeyan, AP(SG)/Mech secured Top 2% position with Gold in an online course on "**Metal Additive Manufacturing**" offered by IIT Kanpur via NPTEL.
- Mr. R. Arun Kumar, AP(SG)/Mech is selected as an "**Honorary Member**" of the IUCEE Consortium
- Dr. P. Sureshkumar, Prof./Mech is awarded with "**NPTEL Believer**" and "**NPTEL Enthusiasts**" during July- Dec., 2024 semester.
- Mr. S. Valai Ganesh, AP(SG)/Mech is awarded with "**NPTEL Believer**" during July-Dec., 2024 semester.
- Mr. R. Arun Kumar, AP(SG)/Mech is awarded with "**NPTEL Evangelist**" during July-Dec., 2024 semester.
- The Department of Mechanical Engineering granted 14th Patent titled "**Fish Desiccation Device**" on 25 Feb., 2025. The inventors are Mr. M. Gopinath, Mr. S. Harish Kumar, Mr. S. Manivignesh, Mr. S. H. Mohamed Yusuff, Alumni Batch 2015-2019 Mr. S. Valai

Ganesh, AP(SG)/Mech, Dr. S. Rajakarunakaran, Prof.&Head/Mech, Mr. T. Selva Sundar and Dr. S. Godwin Barnabas

- Dr. V. Sivakumar, Prof./Mech Secured Topper 1% in NPTEL online course "**Waste to Energy Conversion**" offered by IIT Roorkee during Jan- Mar., 2025.
- Dr. M. Lakshmanan, ASP-II/Mech received Appreciation certificate for acted as a Mentor in TANCAM's HACKATHON for Tamilnadu Women in Science and Engineering (TNWISE 2025) on the theme of "**Smart Assistive Technology**" conducted by Tamilnadu Centre of Excellence for Advanced Manufacturing (TANCAM), Chennai in Association with Rajalakshmi Engineering College, Chennai on 11 Apr., 2025.
- Dr. S. Maharajan, ASP-I/Mech secured Top 5% position in an online course on "**Advanced Machining Processes**" offered by IIT Kanpur via NPTEL during Jan., - Apr., 2025
- Dr. V. Sivakumar, Prof./Mech secured Top 5% position in an online course on "**Non-conventional energy Resources**" offered by IIT Madras via NPTEL during Jan., - Apr., 2025
- Dr. S. Maharajan, ASP-I/Mech secured Topper position in an online course on "**Fundamentals of Surface Engineering: Mechanisms, Processes and Characteristics**" offered by IIT Roorkee via NPTEL during Jan., - Apr., 2025
- Dr. P. Sureshkumar, Prof./Mech is awarded with "**NPTEL Discipline Star**" during Jan.,- Apr., 2025 semester.
- Mr. S. Valai Ganesh, AP(SG)/Mech is awarded with "**NPTEL Believer**" and "**NPTEL Discipline Star**" during Jan.,- Apr., 2025 semester.
- The Department of Mechanical Engineering received a project grant of Rs.70,000 from MSME 4.0 Youth Scheme for the project "**Domestic Refrigerator with Vacuum Insulated Wall (IDEATN02623)**". Mr. R. Muthukumar R, Ms. H. Nivetha, Mr. R. Sri Sabari Venkatesh and Mr. B. Vignesh, Batch of 2021 - 2025 submitted the idea along with mentor by Mr. M. Sivagaminathan alias Balaji, AP(SG)/Mech.

DC MEMBER

- Dr. V. Sivakumar, Prof./Mech attended the "**Doctoral Committee Meeting**" to confirm the provisional Anna University Ph.D. registration of Mr. P. T. Dineshkumar (21142997158) on 29 Nov., 2024 at 10.00 a.m. through online mode.



- Dr. V. Sivakumar, Prof./Mech. attended the **annual Doctoral Committee Meeting** for the part time Anna University research scholar Mr.A.Sankara Narayana Murthy (20142891117) on 06.12.2024 at 11.30 a.m. through online mode.

